TU257 Information Systems

Assignment 1

PART 1 Conceptual Design (20%)

Business Requirements

A public rental bike company (OurCity Bikes) has hired you to design their database. Their initial requirements are:

- The Company owns a number of bicycles and stations across the city.
- Each bicycle and station are identified using an ID.
- Each station has a number of parking slots which can be available or unavailable (in use). The stations also have characteristics associated with them, for example whether you can pay with card or not.
- Users of the scheme have to register in order to obtain their bike card.
- The company needs to identify each trip across the city to calculate charges (Charges apply after 30 minutes of continuous use). User will have stored their payment details at registration so the charges can be processed automatically.

Note: You are free to make any assumptions about the information given. Any assumptions made must be included in your answer.

Create a schema for this company. Include a description of the design process with entities, relationships, cardinalities and attributes. Provide also the Rough ER and Fully Attributed ER diagrams as well as any assumptions taken. A final reflection on the work done including limitations of the proposed design and possible improvements should be included too. See marking rubric in page 3 for more details.

PART 2 Logical Design (10%)

The form below lists the student record information for a typical student at a third level college. **Describe and illustrate the steps of normalization by converting the form below from Un-normalized Form (0NF) to Third Normal Form (3NF).**

Student Details:			Date: 10/12/2023
Student No:	99123456		
Student Name:	Joe Bloggs		
Course Enrolled:	CBEM		
Mode of Study:	Full Time		
Student Advisor No:	A124		
Student Advisor Name:	A. N. O'Reilly		
Subject Details:			
Subject Number	Subject Name	Year/Semester	Grade
COMP 335	Databases	2022/1	A
COMP 304	Operating Systems	2022/2	B

Note: You are free to make any assumptions about the information given. Any assumptions made must be included in your answer.

a) Un-normalised form (i.e. 0NF)
b) Decomposition of relation into 1NF
c) Decomposition of relation into 2NF
d) Decomposition of relation into 3NF

Submission

<u>Deadline week 8.</u>

Submit a report including solutions for parts 1 and 2. Format for the report can be written or video/voice over content (diagrams, etc..).

Marking Rubric

PART 1	**Marks Basic Solution**	**Marks Complex Solution**
Assumptions	2	2
Entities Identification	2	3
Relationships and cardinality	2	3
Attributes	2	3
Rough ER Diagram	2	3
Fully Attributed ER Diagram	2	4
Reflection Limitations and possible improvements	1	2
PART 2		
UNF	1	1
1NF	1	1
2NF	3	3
3NF	3	3
Normalisation Assumptions	2	2
Total	**23**	**30**